

TECHNICAL DESIGN · CODEX CLI AGENT RUNTIME

Agent Agent
Prompt · Tools · Skills · Harness

□□	v1.0	
□□	2026-07-17	
□□□	Codex CLI	ephemeral Skill
□□	DOCX	equipment_deep_research

2026-07-17

JS

《JS》equipment_deep_research

1.

- | | | | | |
|---------------|---|---|--|--------|
| Agent | “ | → | ”typed handoff | Prompt |
| 1. Agent | / | | 3—5 | |
| 2. Agent | | | 2—3 | 8 |
| 3. | | | Packet EvidenceCard ID | |
| 4. DeepSearch | “ | ” | search_sources → fetch_page → create_evidence_card | |
| 5. | | | | |
| 6. Harness | | | Trace | |

2.



Typed handoff · SavePoint · Recall from earliest failed node

1 typed handoff

2.1

- Agent
- Agent
- Agent Agent typed Packet

3.

3.1

- fact
- assessment
- hypothesis
- assumption
- unknown
- conflict

3.2 EvidenceCard

```
{
  "evidence_id": "ev-...",
  "claim": " ",
  "source_type": "government|military|think_tank|industry|academic|news",
  "source_date": "YYYY-MM-DD",
  "body_location": " ",
  "support_level": "direct|partial|context|counter",
  "freshness": "current|aging|historical",
  "confidence": 0.0
}
```

3.3

-
-
- COA
-

3.4 Codex CLI

Agent	Harness	--ephemeral	Codex CLI
7. web_discovery	Codex Web Search		EvidenceCard
8. evidence_analysis	Web Search	assignment	typed handoff Schema
JSON			

```
[
  {
    "role": "system",
    "content": " $js-equipment-agent-runtime Codex CLI Agent agent_runtime task_input Skill
    Harness Codex JSON Markdown "
  },
]
```

```
{
  "role": "user",
  "content": {
    "agent_runtime": {
      "agent_id": "international_situation|combat_scenario",
      "phase": "web_discovery|evidence_analysis",
      "active_skills": [],
      "tools": [],
      "harness": {},
      "quality_gates": [],
      "stop_conditions": [],
      "output_contract": {}
    },
    "task_input": {}
  }
}
```

Skill

- Agent js-equipment-agent-runtime/SKILL.md
- agent_id=international_situation references/international_situation.md
- agent_id=combat_scenario references/combat_scenario.md
-

Codex CLI	Harness Tool	agent_runtime.tools	Harness	Schema
EvidenceCard	Trace	“ ”		

4. Agent

4.1

Agent ID	international_situation
Harness Profile	strategic_research_v1
	/
N	3 2–5 5
	Chief Orchestrator RecallRequest
	EvidenceCard StrategicAssessment BaselineFindingPacket WorkingCheckpoint
	“ ”

4.2

[illegible]

```

    "public_sources_only": true,
    "major_claim_double_source": true,
    "counter_evidence_required": true
  },
  "hard_constraints": [
    "recall_request": null
  ]
}

```

4.3

alternative_hypotheses

scenario_drivers

```

{
  "situation_assessment": " ",
  "threat_assessment": " ",
  "strategic_pattern": " ",
  "opponent_moves": [
    " "
  ],
  "warning_indicators": [
    {
      "indicator_id": "wi-01",
      "type": "leading|coincident|lagging",
      "observable": " ",
      "source_channel": " ",
      "supports_hypotheses": ["bh-01"],
      "confidence": 0.0
    }
  ],
  "alternative_hypotheses": [
    {
      "hypothesis_id": "bh-01",
      "name": " ",
      "status": "candidate|preferred|alternative|watchlist",
      "time_horizon": "near|mid|long",
      "actors": [
        " "
      ],
      "geography": [
        " "
      ],
      "drivers": [
        " "
      ],
      "trigger_conditions": [
        " "
      ],
      "event_chain": [
        " "
      ],
      "de_escalation_conditions": [
        " "
      ],
      "counter_hypothesis": " ",
      "falsification_signals": [
        " "
      ],
      "evidence_ids": ["ev-..."],
      "counter_evidence_ids": ["ev-..."],
      "confidence": 0.0,
      "ranking_score": 0.0
    }
  ],
  "scenario_drivers": [
    {
      "hypothesis_id": "bh-01",
      "mission_pressure": " ",
      "force_constraints": [
        " "
      ],
      "regional_constraints": [
        " "
      ],
      "observable_triggers": [
        " "
      ],
      "critical_uncertainties": [
        " "
      ]
    }
  ]
}

```

4.4 System Prompt

“ Agent” JS



4.5 Skills

Skill A strategic_osint

-
-
- SOP → → → → EvidenceCard
- 3
-

Skill B threat_forecasting

-
- SOP → → / → →
-
-

Skill C force_posture_tracking

-
- SOP → → → / / →
-
-

4.6 Tools

Tool				
search_sources		query source_types time_range language	SearchLead[]	
fetch_page		URL/lead_id	SourceMaterial	

Tool				
create_evidence_card		claim excerpt location assessment	EvidenceCard	
register_event_timeline		actor event date location evidence_ids	StrategicAssessment	
compare_actor_positions		actors issues evidence_ids	position matrix	
register_warning_indicator		indicator type threshold sources	warning indicator	
test_competing_hypothesis	/	hypotheses support counter_evidence	comparison result	

4.7 Harness Profile

```

profile_id: strategic_research_v1
context_policy:
  token_budget: 6500
  hide_other_agent_raw_sessions: true
budget:
  max_turns: 8
  max_searches: 16
  max_tool_calls: 32
  max_tokens: 6500
evidence_policy:
  double_source_major_claims: true
  require_counter_evidence: true
  require_time_horizon: true
checkpoint_policy:
  after_each_turn: true
  after_required_artifact: true
stop_conditions:
  - 2-5
  -
  -
  -
recovery_policy:
  resume_from: earliest_incomplete_artifact
  retry_limit: 2

```

5. Agent

5.1

Agent ID	combat_scenario
Harness Profile	scenario_research_v1
M	2 1-3 8
	typed Packet
	EvidenceCard ScenarioModel BaselineFindingPacket WorkingCheckpoint

5.2

```
{
  "task_id": "string",
  "research_route": "new_winning_mechanism|traditional_gap|war_case_learning",
  "task_boundary": " ",
  "background_hypotheses": ["StrategicAssessment.alternative_hypotheses"],
  "scenario_drivers": ["StrategicAssessment.scenario_drivers"],
  "expert_fixed_background": null,
  "required_dimensions": ["opponent", "region", "mission_type", "intensity", "time_window"],
  "scenario_horizons": [" ", " ", " ", " ", " ", " ", " "],
  "hard_constraints": [" "],
  "recall_request": null
}
```

5.3

```
{
  "scenario_framework": {
    "background_hypothesis_ids": ["bh-01"],
    "participants": [" "],
    "region": [" "],
    "mission_objectives": [" "],
    "phases": [" "],
    "entry_conditions": [" "],
    "termination_conditions": [" "],
    "success_criteria": [" "]
  },
  "enemy_coa": [
    {
      "coa_id": "coa-01",
      "class": "most_likely|most_dangerous|alternative",
      "task_level_description": " ",
      "observable_indicators": [" "],
      "critical_dependencies": [" "],
      "evidence_ids": ["ev-..."]
    }
  ],
  "critical_timeline": [
    {
      "phase": " ",
      "decision_window": " ",
      "trigger": " ",
      "consequence": " "
    }
  ],
  "environment_constraints": {
    "terrain": [],
    "weather": [],
    "electromagnetic": [],
    "network": [],
    "space": [],
    "logistics": [],
    "political_legal": []
  },
  "scenario_branches": [
    {
      "scenario_id": "sc-01",
      "background_hypothesis_id": "bh-01",
      "name": " ",
      "opponent": " ",
      "region": " ",
      "mission_type": " ",
      "intensity": "low|medium|high|escalating",
      "time_window": " ",
      "branch_class": "most_likely|most_dangerous|alternative",

```

```
"key_events": ["    "],
"assumptions": ["    "],
"falsification_signals": ["    "],
"evidence_ids": ["ev-..."],
"plausibility_score": 0.0,
"score_breakdown": {
  "evidence_support": 0.0,
  "causal_coherence": 0.0,
  "resource_geography_feasibility": 0.0,
  "trigger_observability": 0.0,
  "branch_distinctness": 0.0
}
},
],
"capability_pressure_points": [
{
  "scenario_id": "sc-01",
  "mission_chain_node": "  |  |  |  |  |  ",
  "pressure": "    ",
  "downstream_question": "    /    Agent    "
}
],
"assumptions": ["    "]
}
```

5.4 System Prompt

“ Agent” JS

Agent typed Packet Agent

1.

2.

3.

4.

5.

6.

7.

2 3 8

ID / ID

“ ”“ ”

COA

Agent

-

“ - - - - - ”

-

-

-

/

-

- ScenarioModel typed Packet

25% 25% 20% 15% 15%

- allowed_tools

- EvidenceCard assumption/unknown

-

-

5.5 Skills

Skill A scenario_engineering

-

- SOP → / / → → / → →
-
-

Skill B `modern_battlespace_analysis`

-
- SOP → → →
-
-

Skill C `adversary_coa_analysis`

-
- SOP / / COA → → →
- COA
- COA

5.6 Tools

Tool				
<code>search_sources</code>		question source_types time_range	SearchLead[]	
<code>fetch_page</code>		URL/lead_id	SourceMaterial	
<code>create_evidence_card</code>		claim excerpt location	EvidenceCard	
<code>build_scenario_graph</code>		actors objectives phases triggers	ScenarioModel	
<code>branch_scenario</code>		parent branch_dimension constraints	scenario branches	
<code>map_critical_window</code>		phase trigger decision_window	critical timeline	
<code>map_environment_constraint</code>		environment mission_chain	constraint map	
<code>stress_test_scenario</code>		scenario stressors hard_constraints	pass/fail + gaps	

5.7 Harness Profile

```

profile_id: scenario_research_v1
context_policy:
  token_budget: 7500
  hide_other_agent_raw_sessions: true
budget:
  max_turns: 8
  max_searches: 10
  max_tool_calls: 28
  max_tokens: 7500
evidence_policy:
  evidence_or_explicit_assumption_per_node: true
  require_counter_evidence: true
checkpoint_policy:
  after_each_turn: true
  after_required_artifact: true
stop_conditions:
  -
  -
  -

```

```
-
-
recovery_policy:
  resume_from: earliest_incomplete_artifact
  retry_limit: 2
```

6. Typed Handoff

Agent	Agent
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23
24	24
25	25
26	26
27	27
28	28
29	29
30	30
31	31
32	32
33	33
34	34
35	35
36	36
37	37
38	38
39	39
40	40
41	41
42	42
43	43
44	44
45	45
46	46
47	47
48	48
49	49
50	50
51	51
52	52
53	53
54	54
55	55
56	56
57	57
58	58
59	59
60	60
61	61
62	62
63	63
64	64
65	65
66	66
67	67
68	68
69	69
70	70
71	71
72	72
73	73
74	74
75	75
76	76
77	77
78	78
79	79
80	80
81	81
82	82
83	83
84	84
85	85
86	86
87	87
88	88
89	89
90	90
91	91
92	92
93	93
94	94
95	95
96	96
97	97
98	98
99	99
100	100

```
{
  "packet_id": "packet-international_situation-...",
  "agent_id": "international_situation",
  "payload_type": "strategic_assessment_v1",
  "alternative_hypotheses": [],
  "scenario_drivers": [],
  "warning_indicators": [],
  "evidence_ids": [],
  "confidence": 0.0,
  "coverage_notes": [],
  "open_questions": [],
  "checkpoint": "checkpoint-id"
}
```

-
- URL
-
- “ ”
- Agent session

7. Harness

7.1

```

phases:
- phase_id: strategic_baseline
  agents: [international_situation]
  commit: savepoint_after_strategic_packet

- phase_id: scenario_modeling
  agents: [combat_scenario]
  wait_for: [savepoint_after_strategic_packet]
  context: [task, evidence_policy, own_checkpoint, recall_request, agent_harness, upstream_handoffs]
  commit: savepoint_after_scenario_packet

- phase_id: downstream_parallel
  agents: [weapon_equipment, operational_employment]
  wait_for: [savepoint_after_scenario_packet]

```

7.2

- $\frac{1}{2}$
- $\frac{1}{2}$

7.3

-
-
-
-
-

7.4 Recall

Agent Agent Recall

```
{
  "source_agent_id": "combat_scenario",
  "target_agent_id": "international_situation",
  "reason": " ",
  "required_data": [ " " ],
  "return_node": "scenario_framework:sc-02",
  "urgency": "normal"
}
```

Recall 2

8.

8.1

```
ranking_score =
  0.30 × evidence_sufficiency
+ 0.25 × causal_coherence
+ 0.20 × trigger_observability
+ 0.15 × task_relevance
+ 0.10 × hypothesis_distinctness
```

- EvidenceCard preferred
- candidate
-

8.2

```
plausibility_score =
  0.25 × evidence_support
+ 0.25 × causal_coherence
+ 0.20 × resource_geography_feasibility
+ 0.15 × trigger_observability
+ 0.15 × branch_distinctness
```

-

-
-
-
- / /

9. Trace

- agent_started
 - search_track_started
 - source_materialized
 - evidence_card_created
 - background_hypothesis_created
 - competing_hypothesis_tested
 - strategic_packet_committed
 - scenario_graph_created
 - scenario_branch_created
 - scenario_scored
 - scenario_stress_tested
 - recall_requested
 - checkpoint_saved
 - agent_completed
 - agent_stopped_with_limitations
-
- -
 - EvidenceCard/
 -
 - Recall
 - Tool

10.

Agent Tools Skills Handoff	configs/equipment_deep_research/agents.yaml
Harness	configs/equipment_deep_research/harness.yaml
Tool	configs/equipment_deep_research/tools.yaml
	src/equipment_deep_research/agents/runtime_profiles.py
Prompt	src/equipment_deep_research/agents/prompts.py
typed Packet	src/equipment_deep_research/domain/models.py
A–H	src/equipment_deep_research/orchestration/blueprints.py configs/equipment_deep_research/routes.yaml

BaselinePromptBuilder

Agent

role_guidance

Prompt

N/M

typed

handoff

Tool Skill

Harness ID

11.

11.1 Agent

- 2–5
- preferred
-
- Packet URL

11.2 Agent

- 1–3
-
-
-
-

11.3

- Agent StrategicAssessment typed Packet Agent session
- Agent SavePoint Agent SavePoint
- Recall [return_node](#)
- /Recall